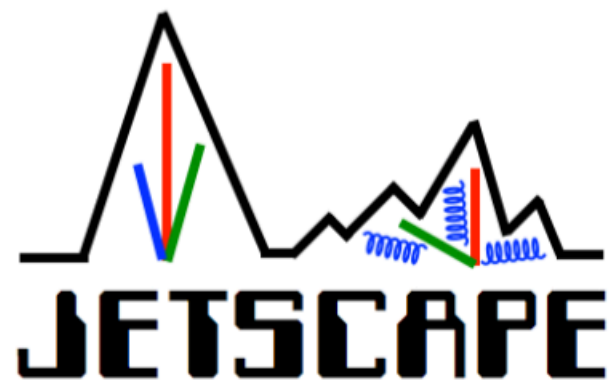




X-SCAPE

Calibrated (3+1)D hydrodynamic medium for ultrarelativistic collision physics

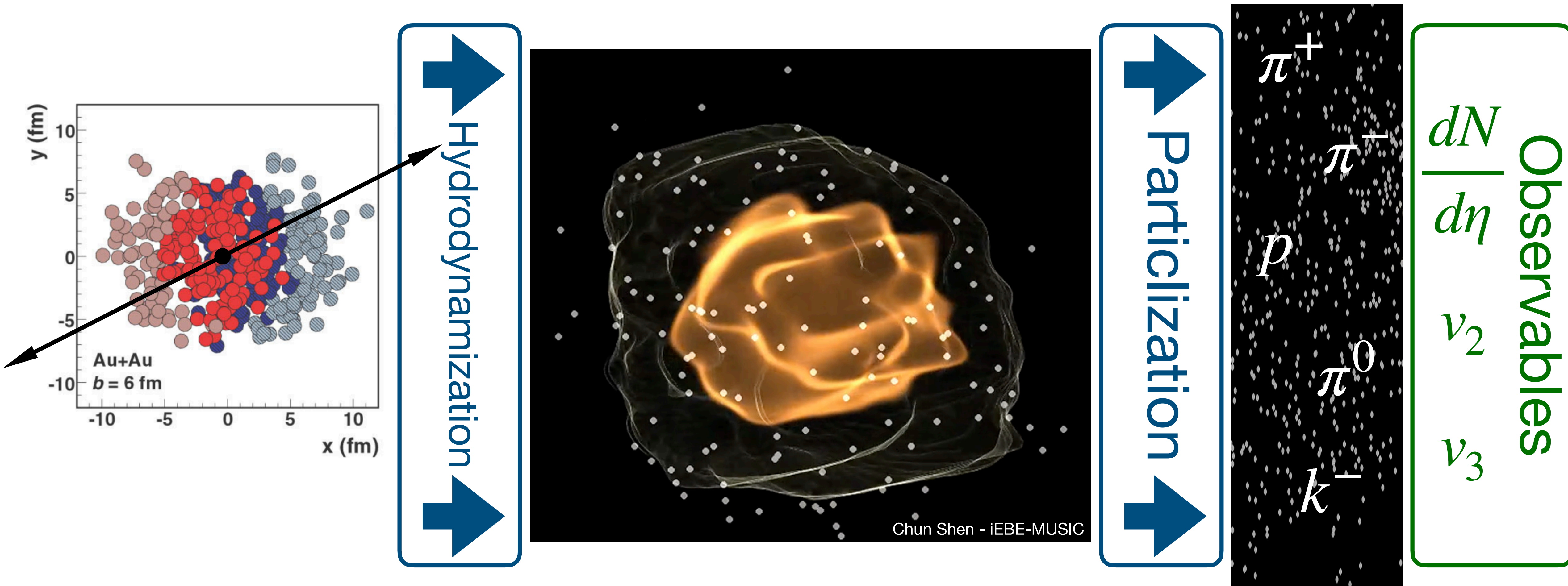
HotJets 2025

***Andi Mankolli on behalf of the
JETSCAPE SIMS working group***

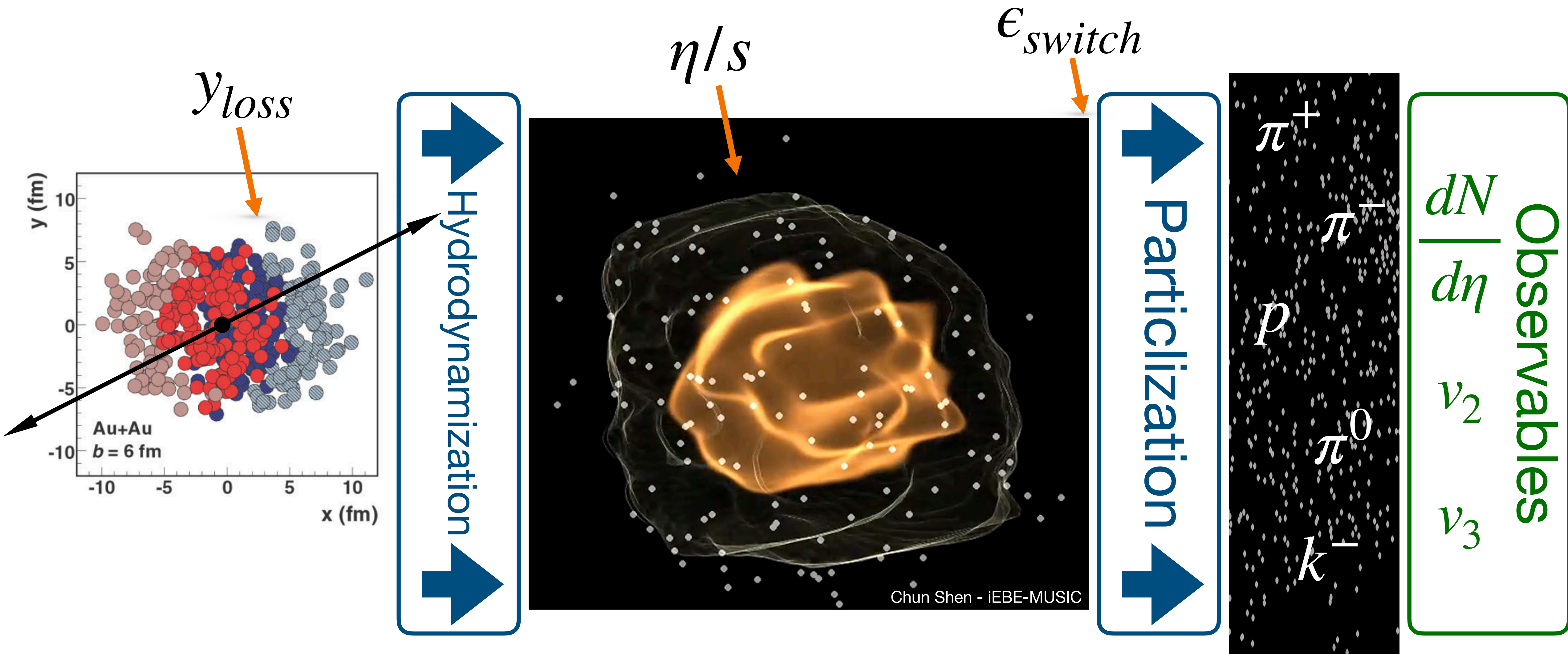
**Advancing the Understanding of High Temperature QCD with Jets
University of Illinois Urbana-Champaign, 8-10 Jan 2025**



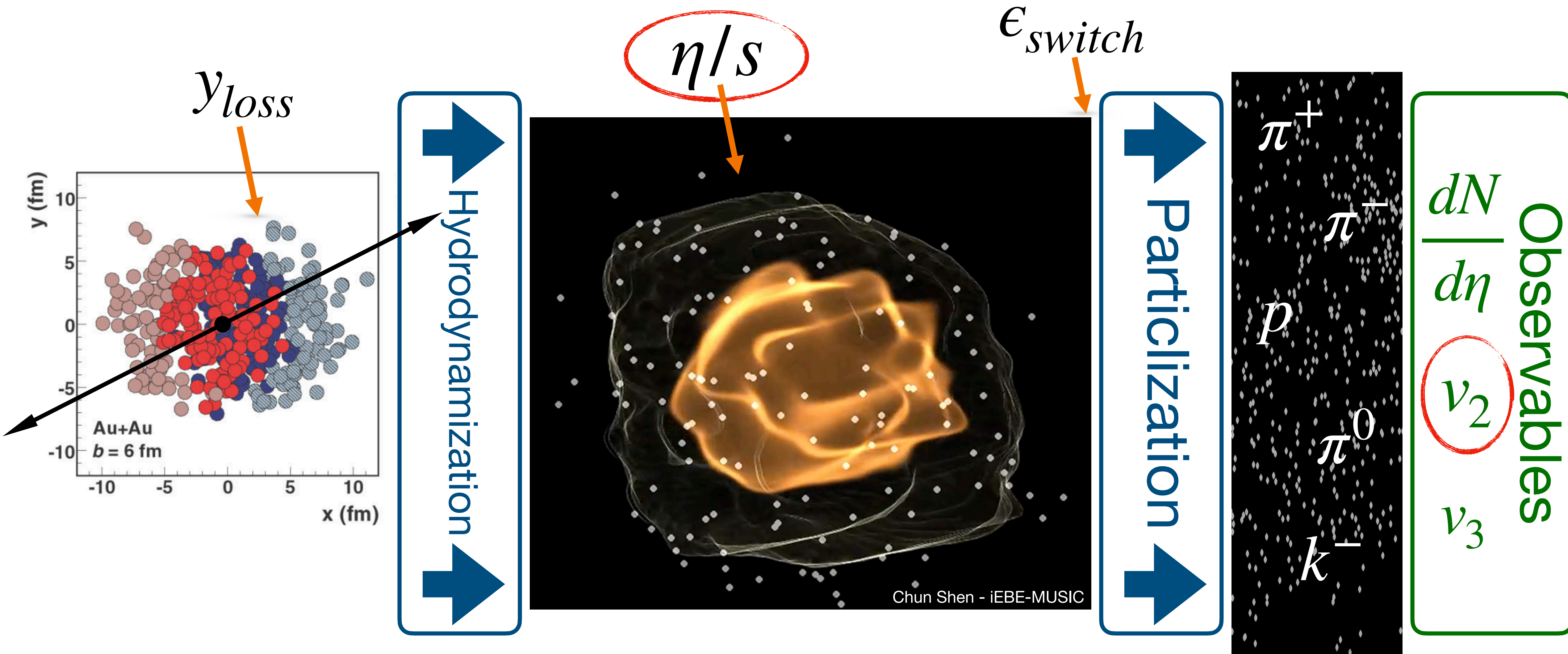
Bulk modeling of QGP



Calibrating bulk models with many observables

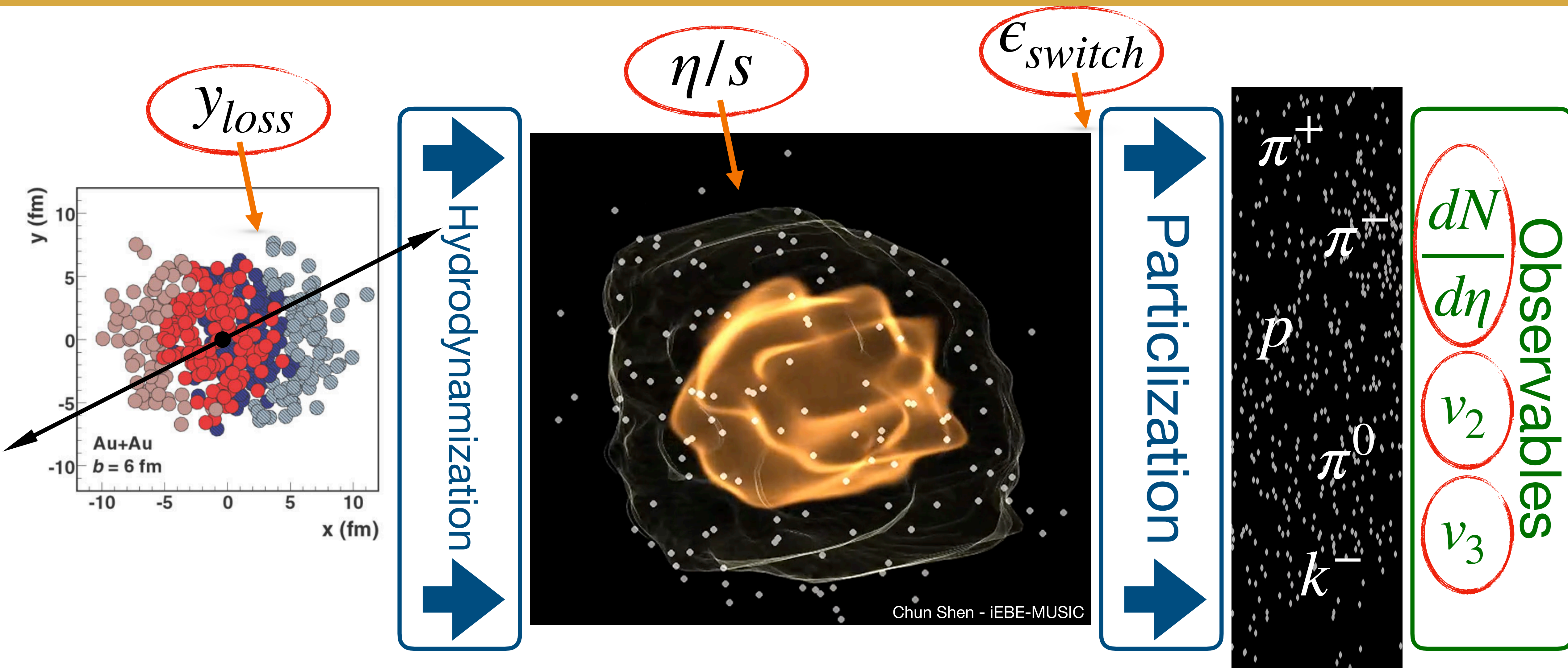


Calibrating bulk models with many observables



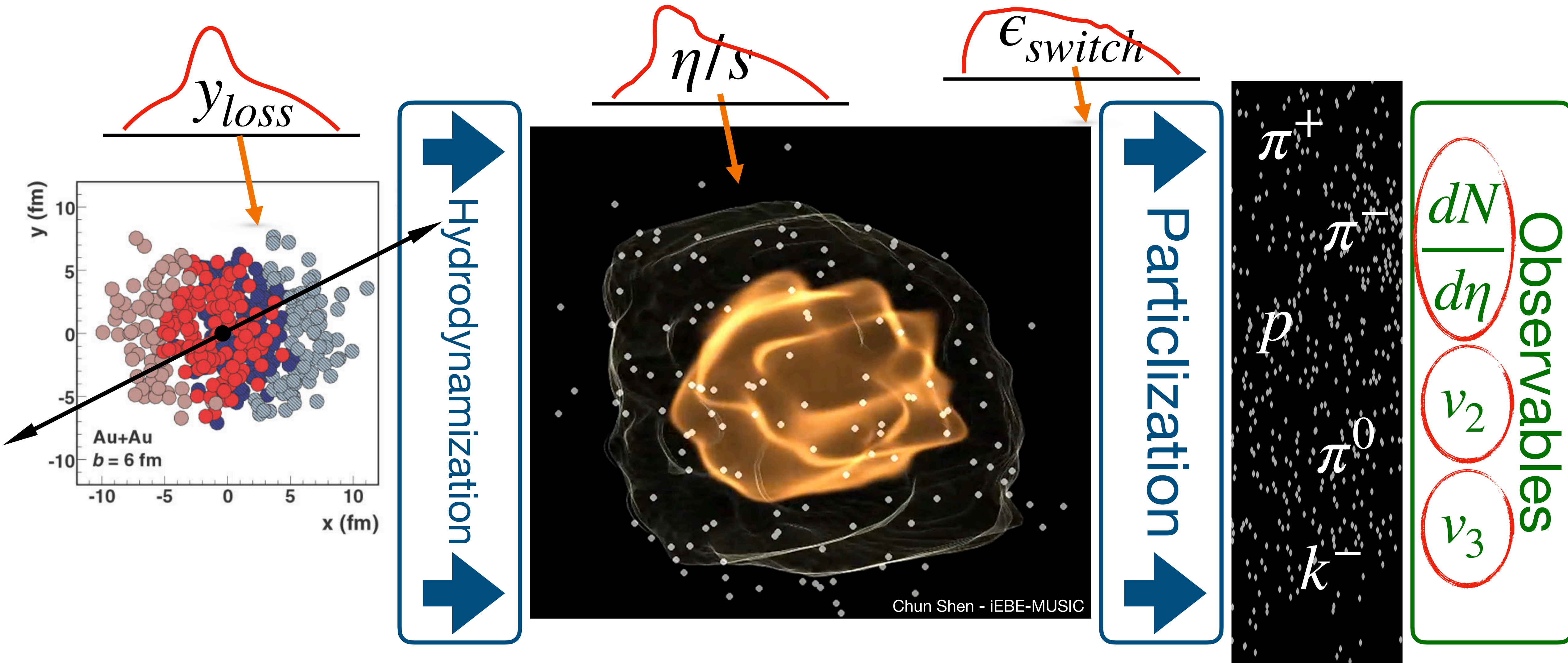
- Targeted fit

Calibrating bulk models with many observables



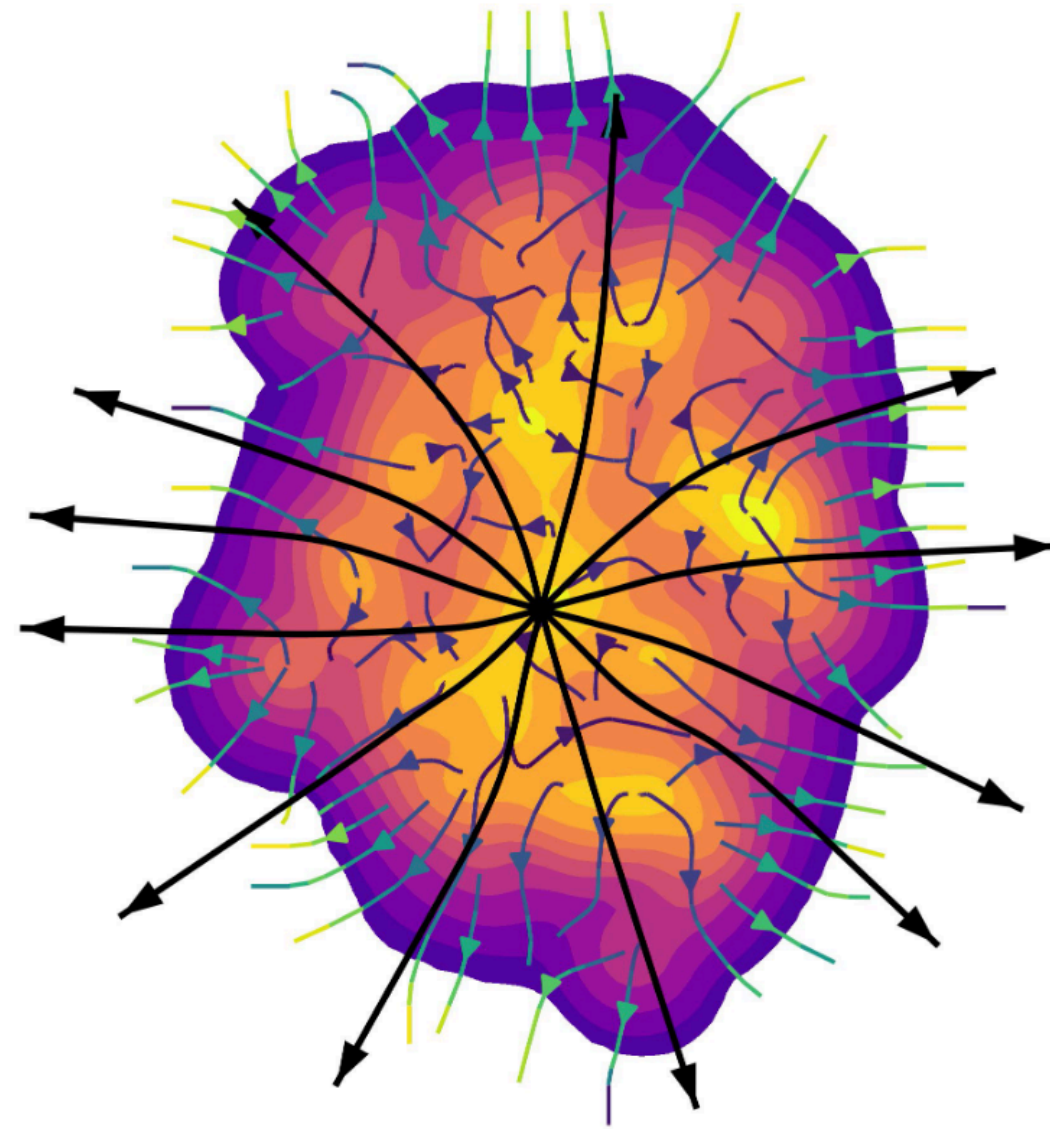
- Global fit

Calibrating bulk models with many observables



- Global Bayesian fit

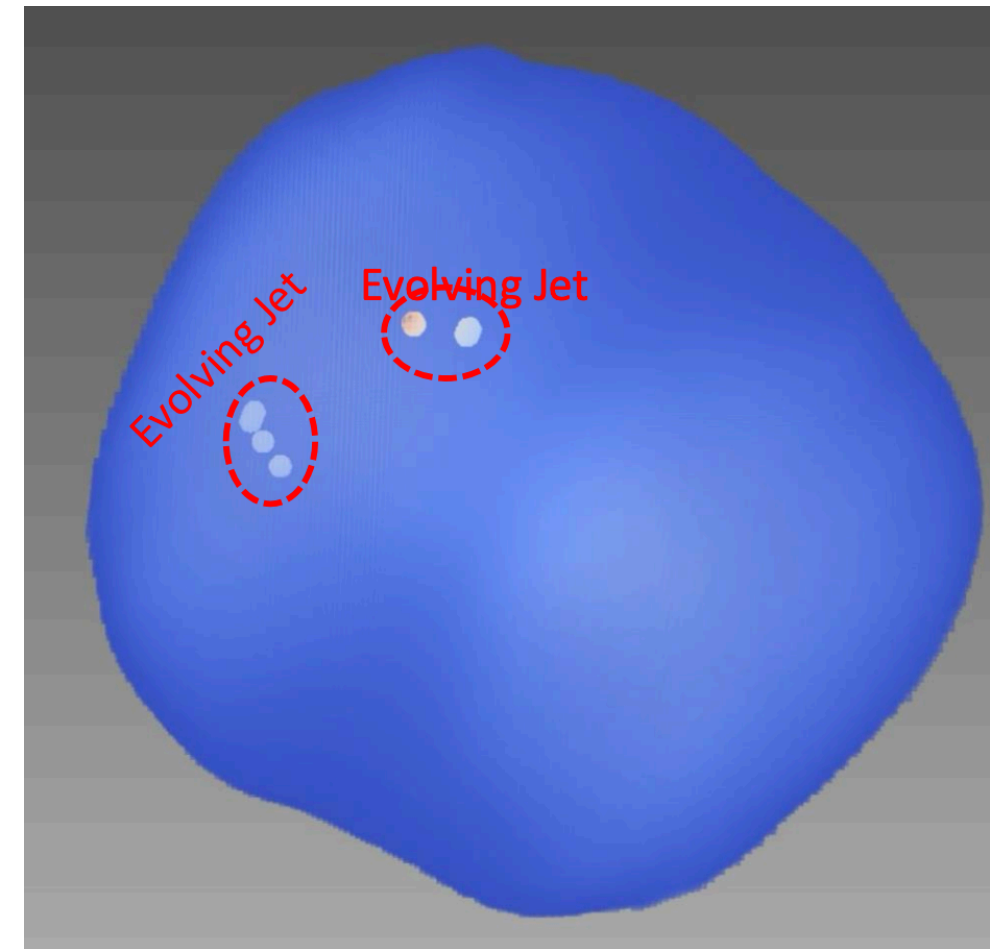
Calibrated medium as a background for hard processes



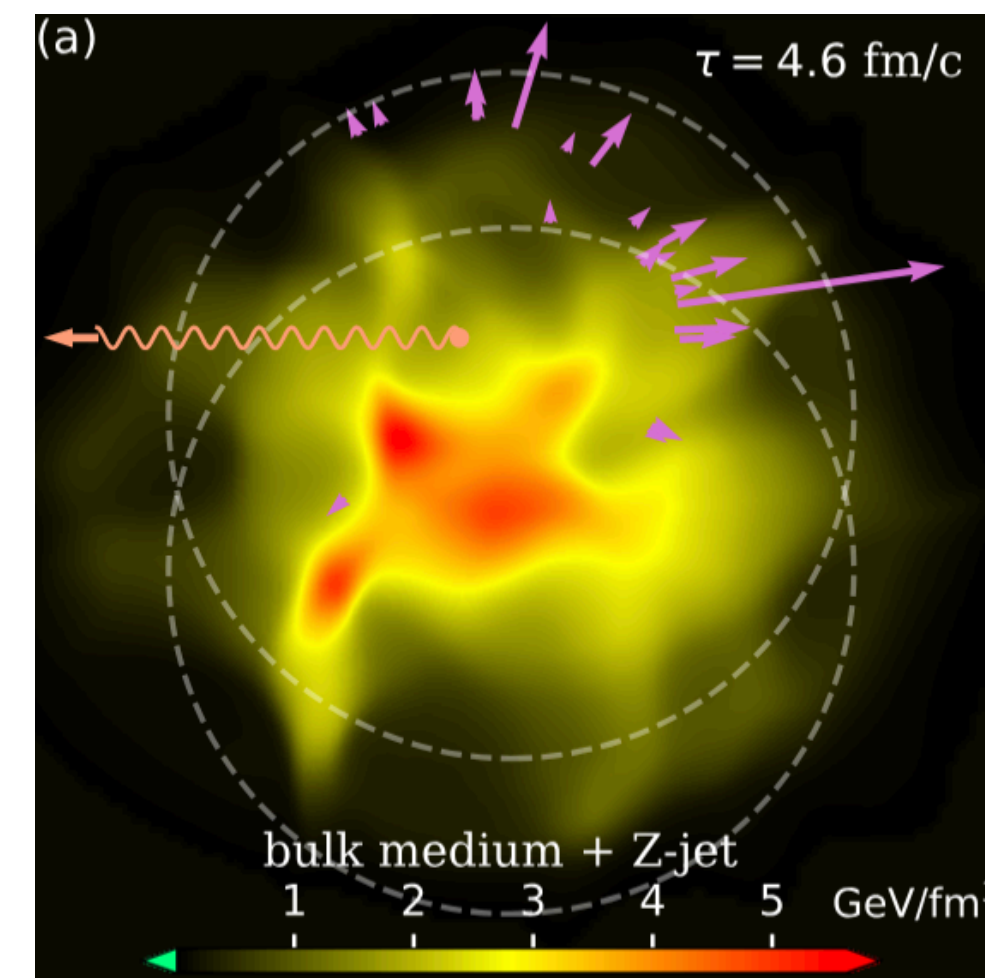
Sievert talk, Wed.

Bahder, Rahman, Sievert, Vitev.

arxiv:2412.05474



Stewart talk, Wed.

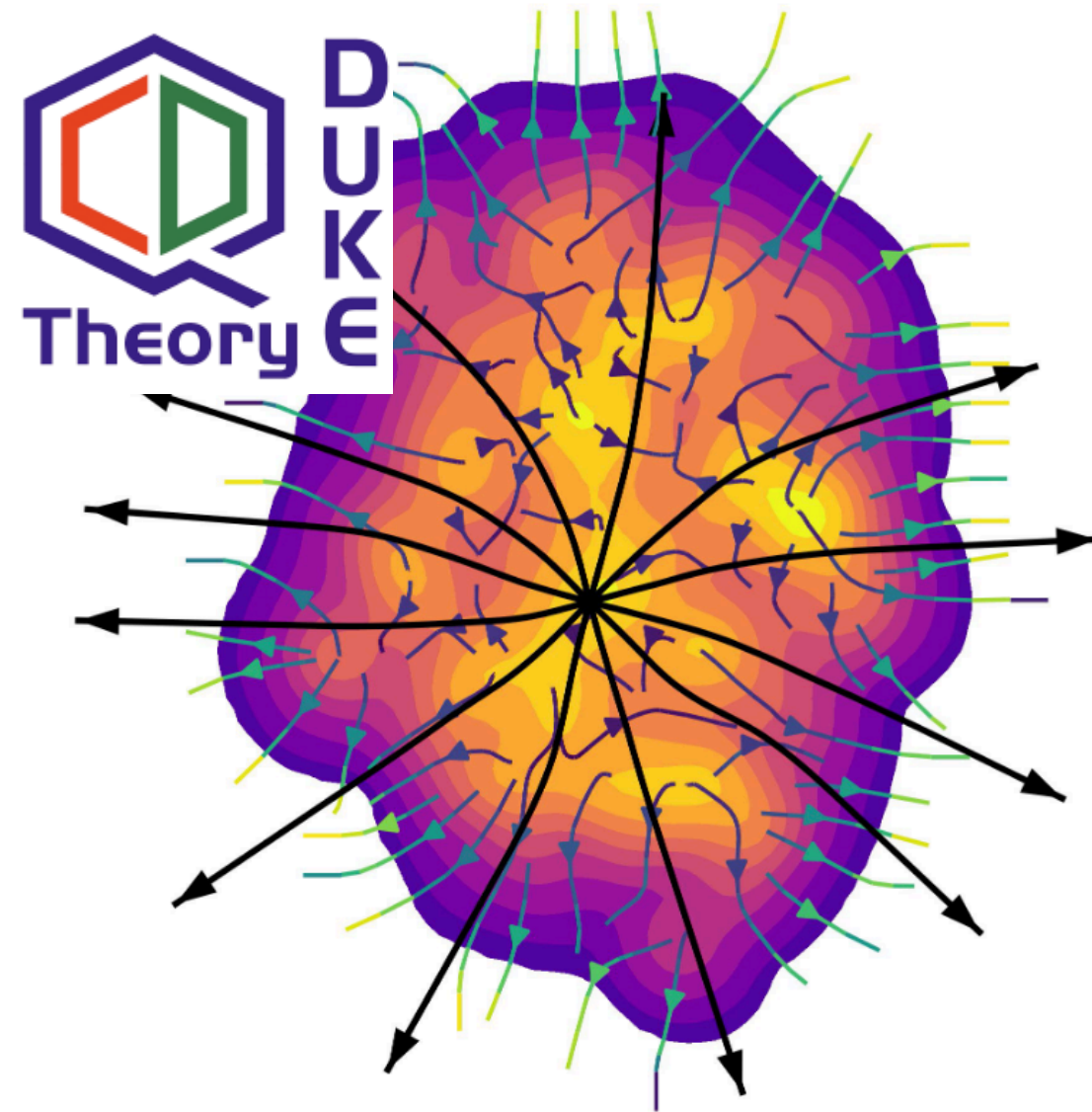


Wang talk, Wed.

Chen, Cao, Luo, Pang, Wang.

PLB777(2018)86

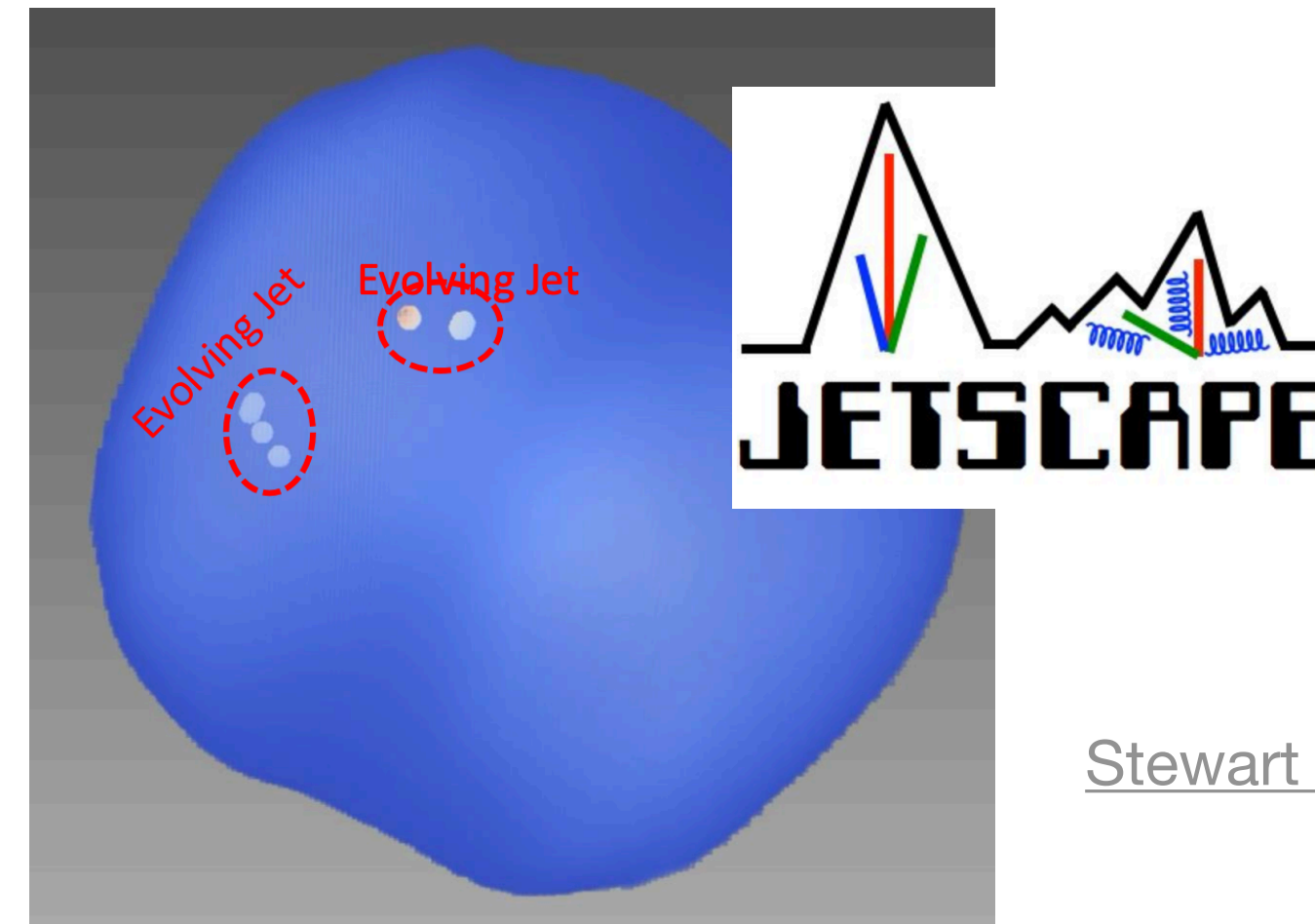
Calibrated medium as a background for hard processes



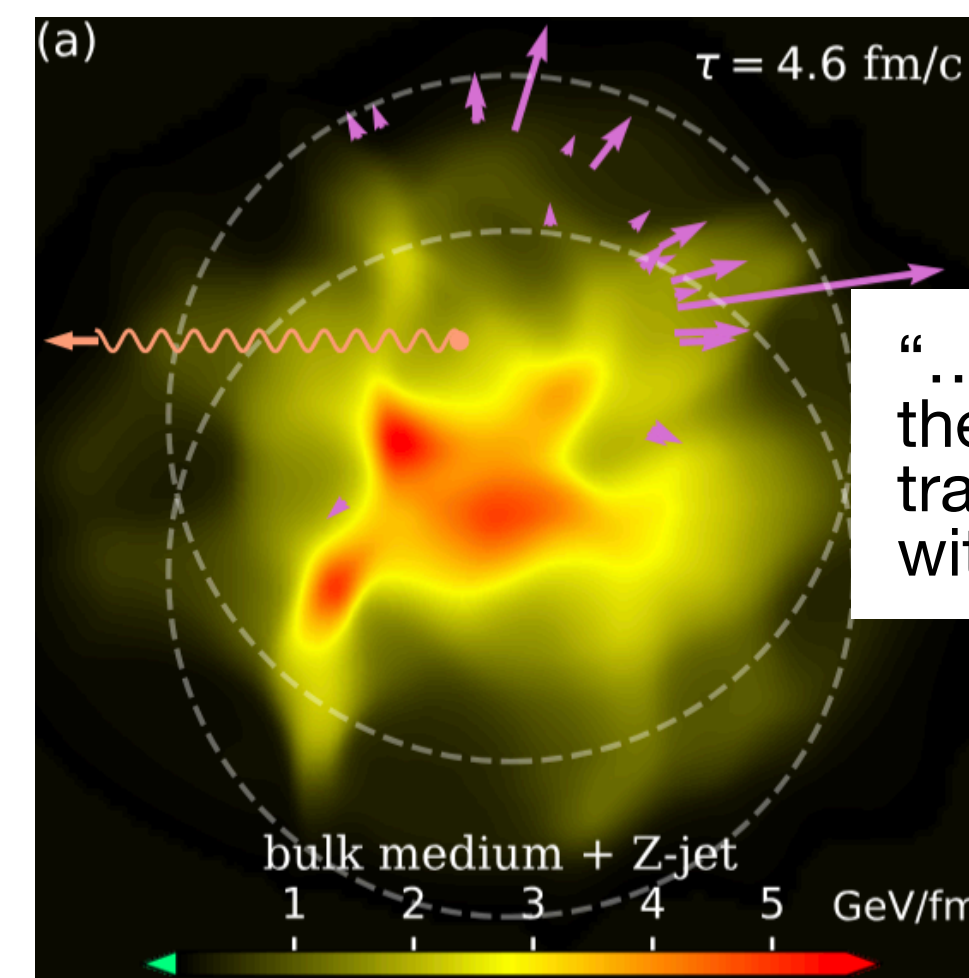
Sievert talk, Wed.

Bahder, Rahman, Sievert, Vitev.

arxiv:2412.05474



Stewart talk, Wed.



“...fitted to reproduce experimental data on the final charged hadron rapidity and transverse momentum distributions...” within Co-LBT

Wang talk, Wed.

Chen, Cao, Luo, Pang, Wang.

PLB777(2018)86

The experimental data for calibration

- Large and small system
- All four RHIC experiments



Au-Au 200 GeV

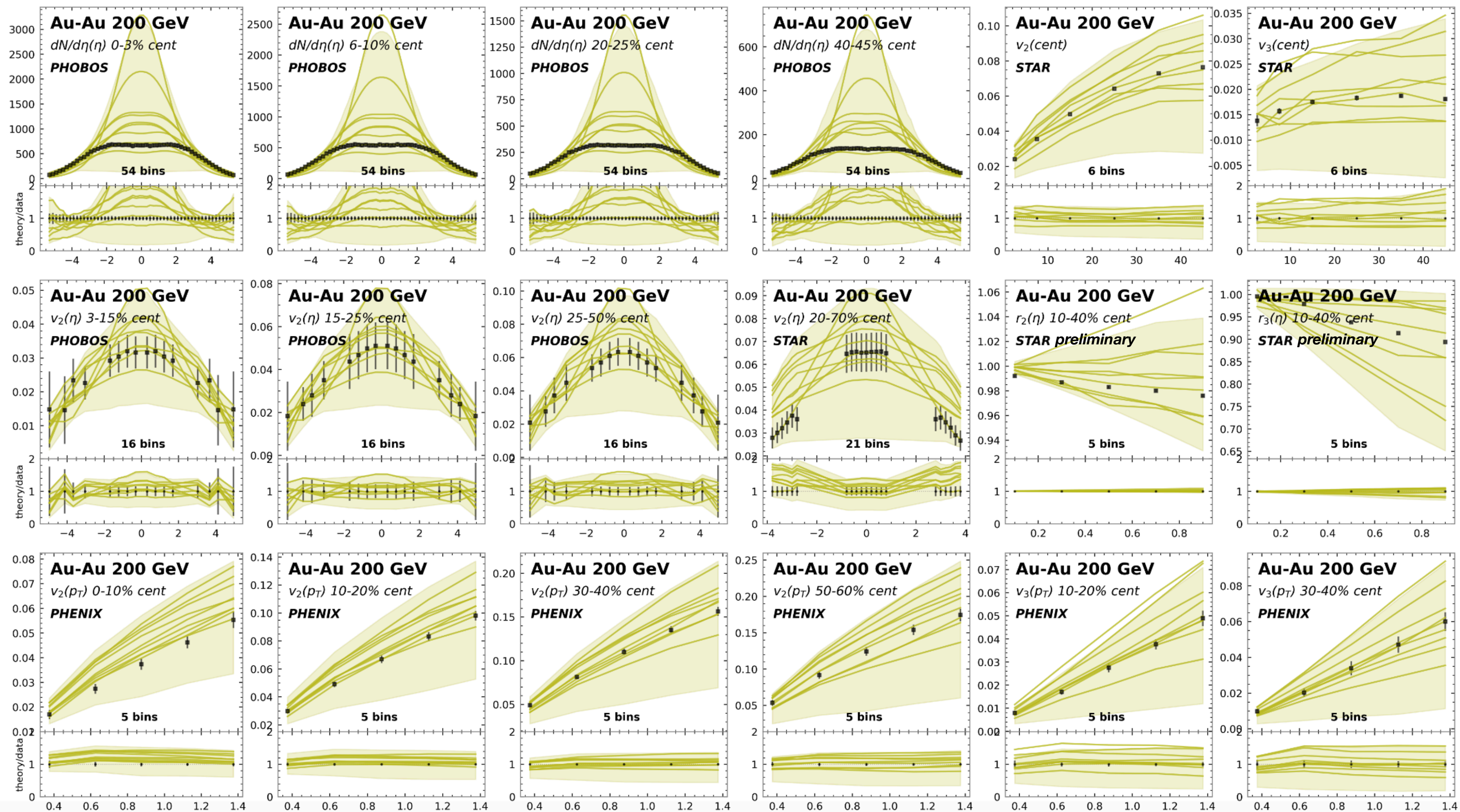
- $dN_{ch}/d\eta(\eta)$ PHOBOS
- $v_2(\eta)$ STAR
- $v_2(\text{cent})$ STAR
- $v_3(\text{cent})$ STAR
- $v_2(p_T)$ PHENIX
- $v_2(p_T)$ STAR
- $v_2(\eta)$ STAR
- $v_2(\eta)$ PHOBOS
- $\langle p_T \rangle \pi, k$ STAR
- $\langle p_T \rangle k, p$ PHENIX

d-Au 200 GeV

- $dN_{ch}/d\eta(\eta)$ PHOBOS
- $dN_{ch}/d\eta(\eta)$ PHENIX
- $v_2(p_T)$ PHENIX
- $v_2(p_T)$ STAR
- $v_2(\eta)$ PHENIX

- Focus on rapidity-dependent observables
- Include p_T -dependent (at low p_T) observables with small uncertainties

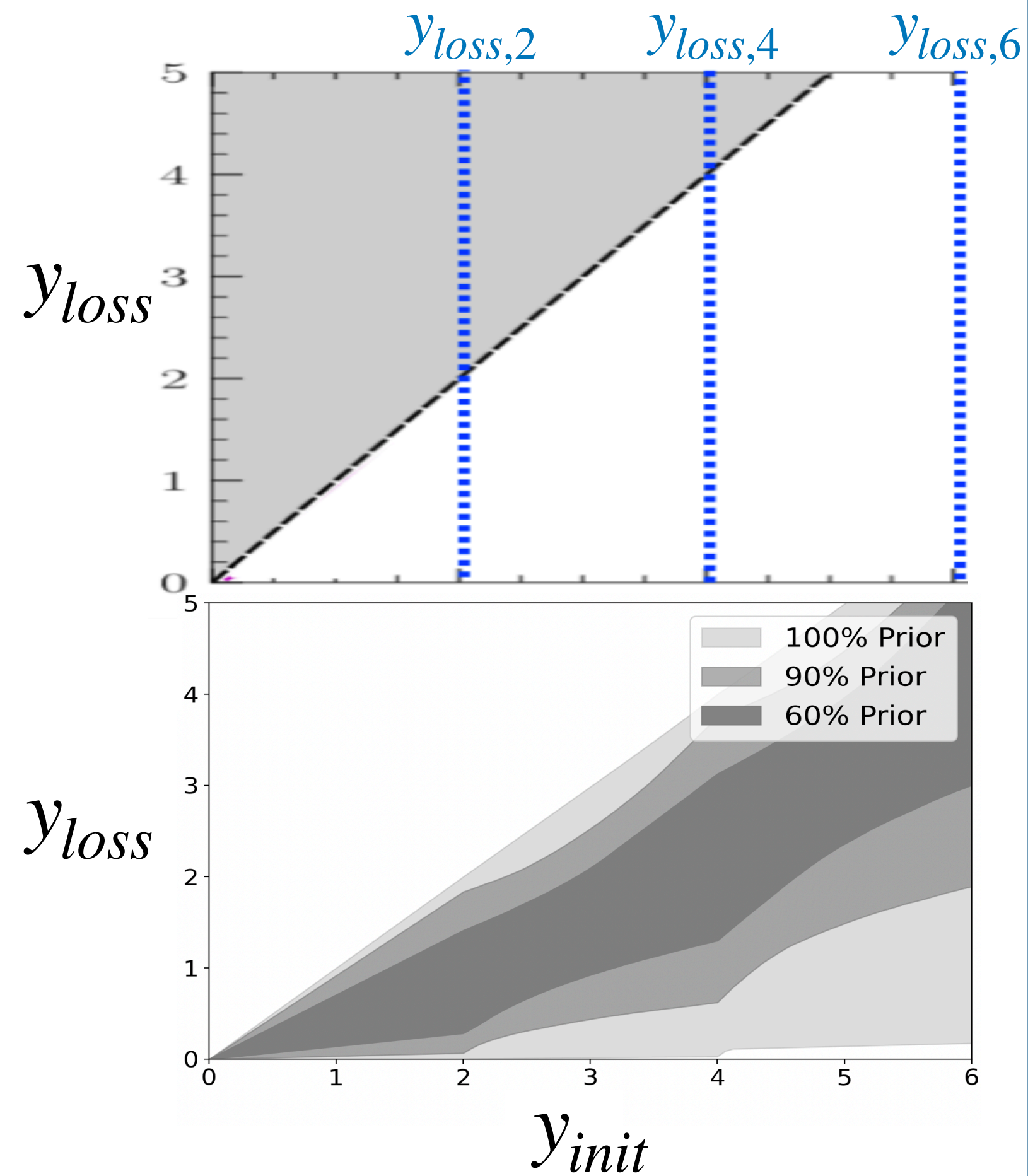
The experimental data and priors



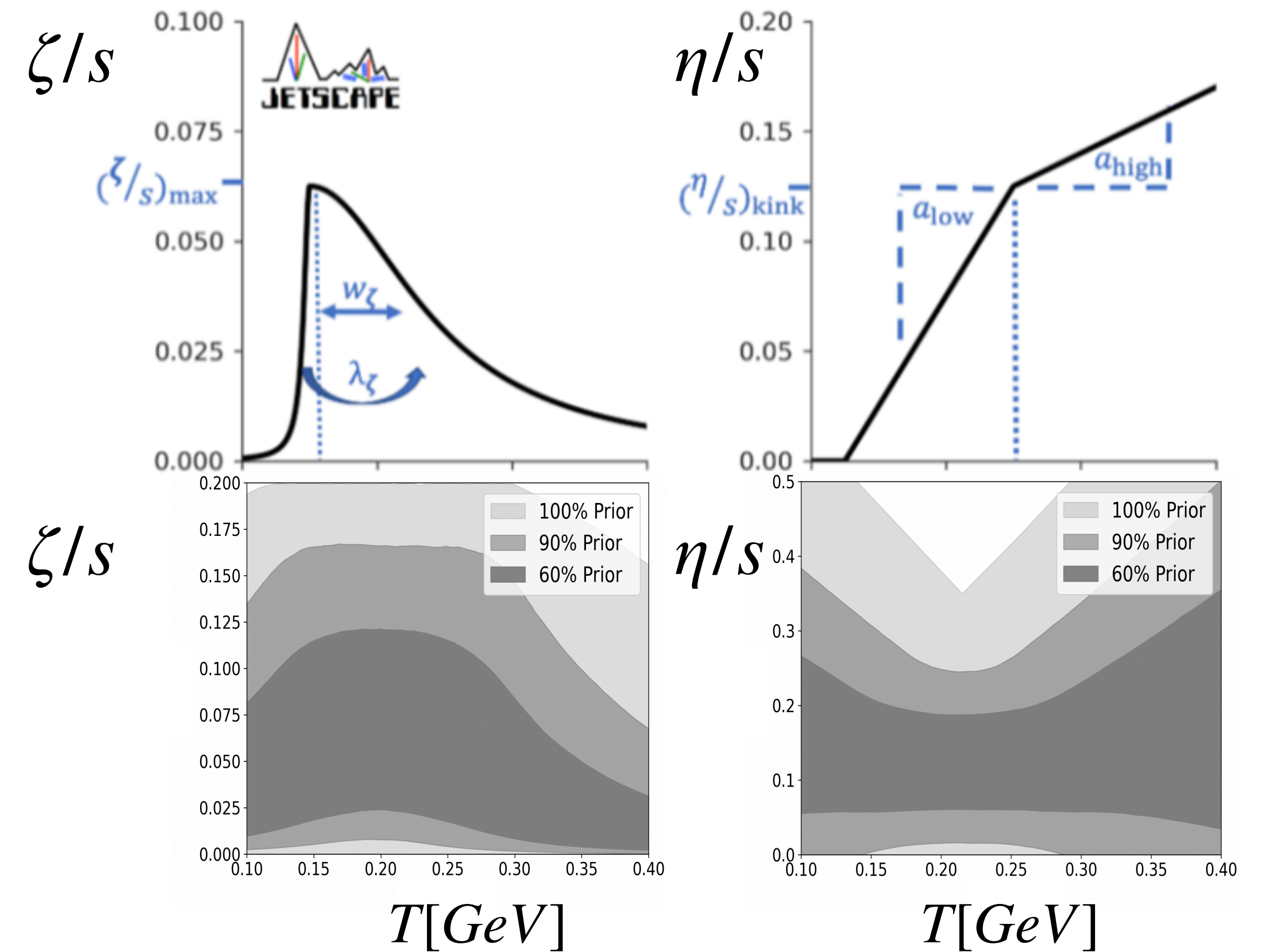
3D model parametrizations

Parametrization
Prior

Initial State



Hydrodynamics



Schenke, Shen, Zhao.

Phys. Rev. C 105, 064905
(2022)

Phys. Rev. C 97, 024907
(2018)



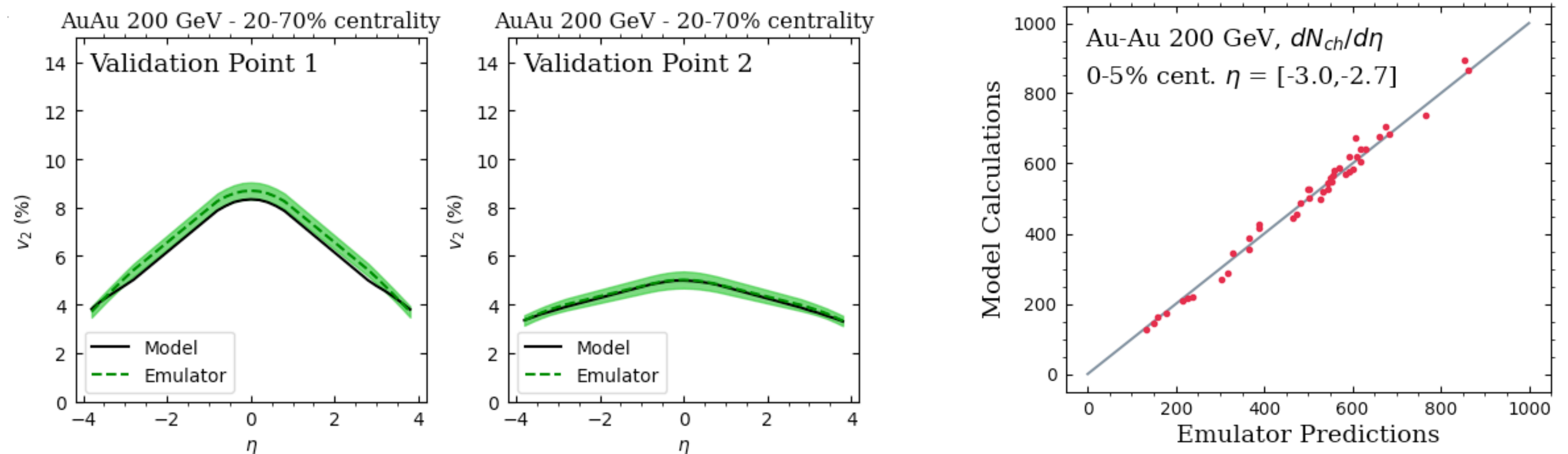
Model emulation and comparison with data

Model simulations

- 20 model parameters
- ~2000 events at 400 parameter points simulated with full model

Model emulation

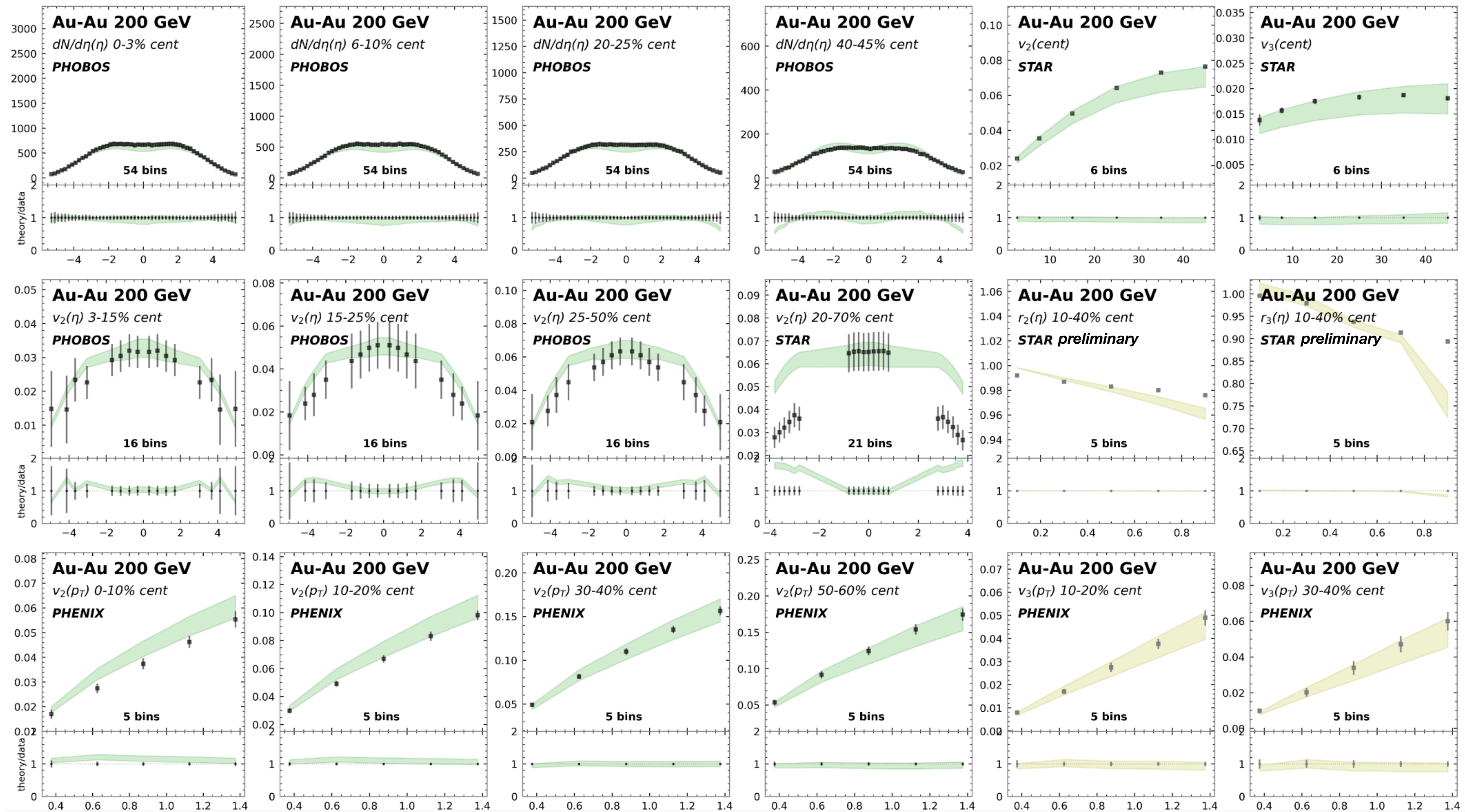
- ~1000 observable bins
- ~10 principal components
- Estimation of uncertainty



Comparison with data

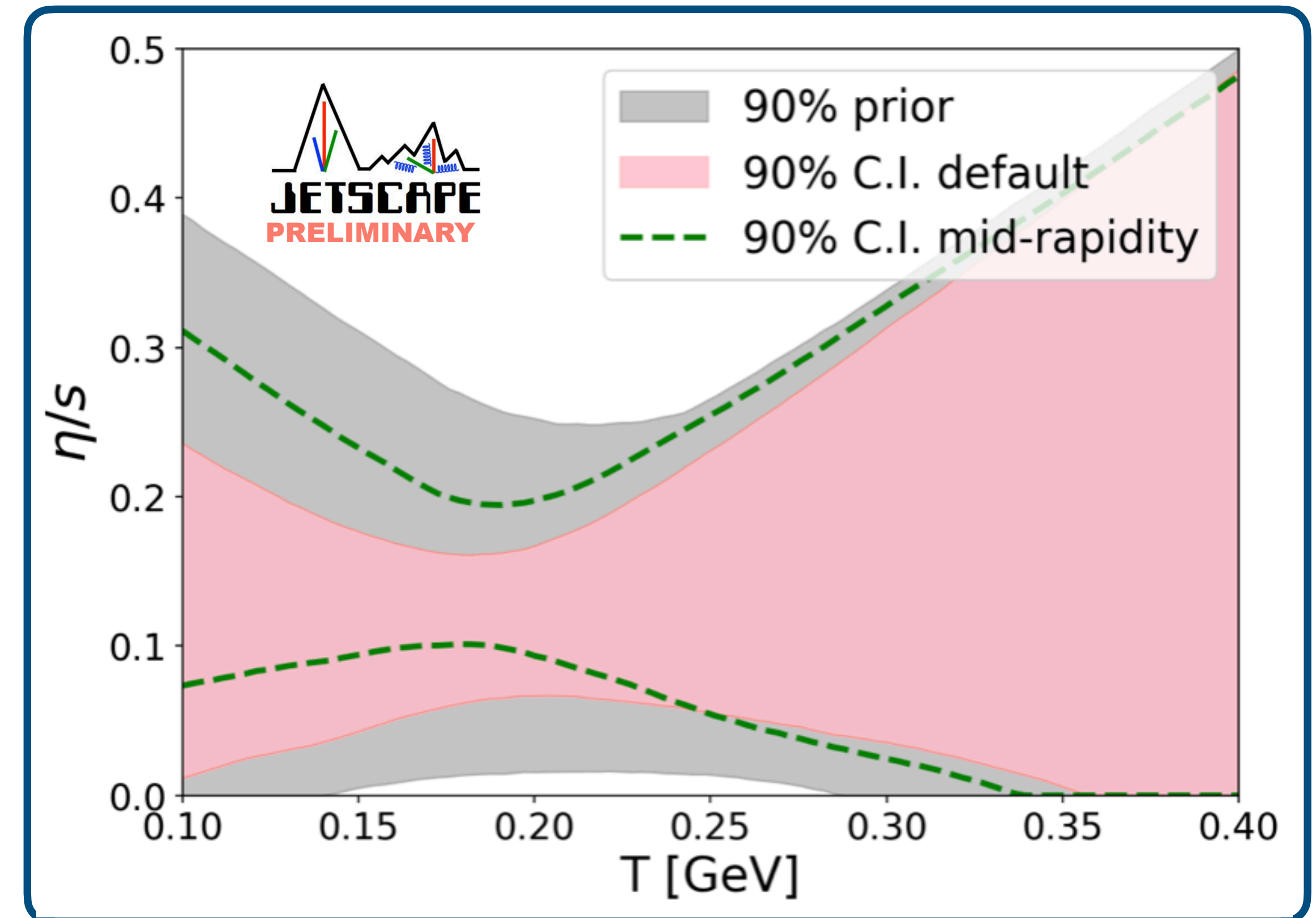
- Likelihood function: $P(Y_{exp}|\theta) \propto \exp \left(-[Y_{exp} - Y_{sim}(\theta)]^T [\Sigma_{exp} + \Sigma_{sim}(\theta)]^{-1} [Y_{exp} - Y_{sim}(\theta)] \right)$

Calibrated description of the data

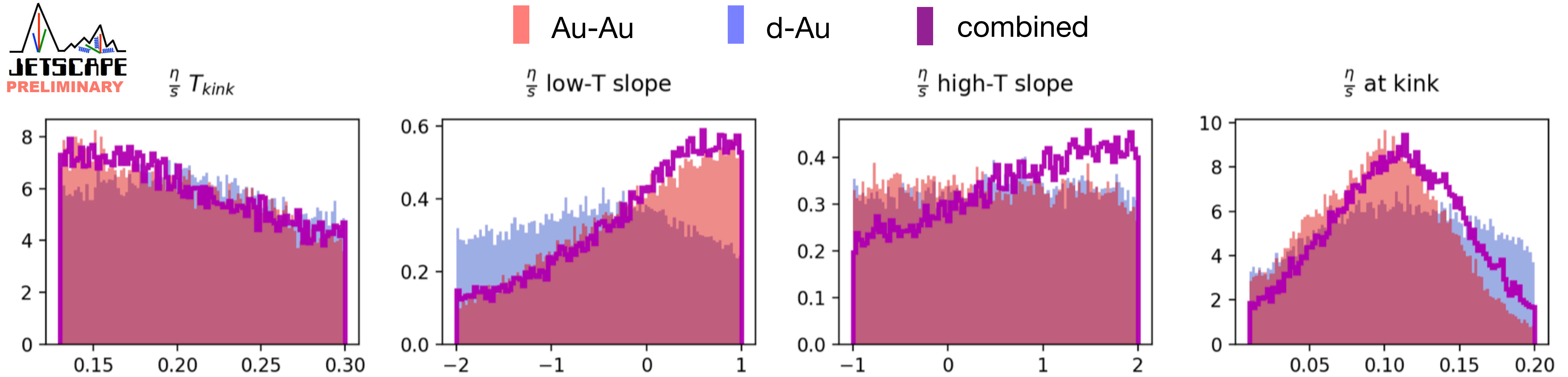


Rapidity-dependent constraints on the shear viscosity

- Mid-rapidity data often includes 3D information
 - Centrality selection
 - Flow measurement reference
- Data with explicit rapidity-dependence reduces the shear viscosity
- Minimal sensitivity at high T



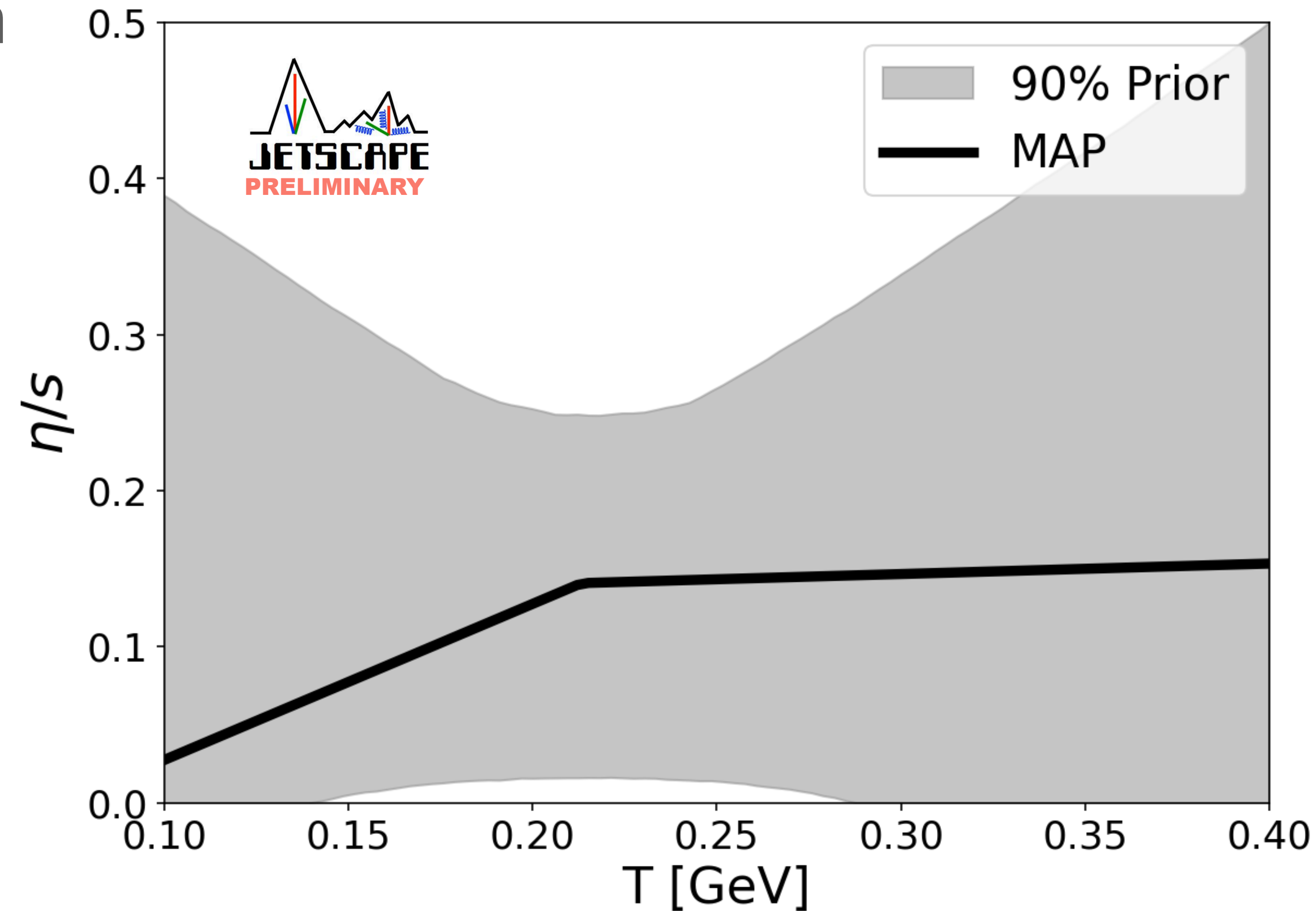
Large and small system constraints on the shear viscosity



- Au-Au and d-Au constraints largely consistent
- Au-Au dominates the fit due to data volume & uncertainties
 - Including d-Au in the fit does not appreciably affect shear viscosity in Au-Au simulations

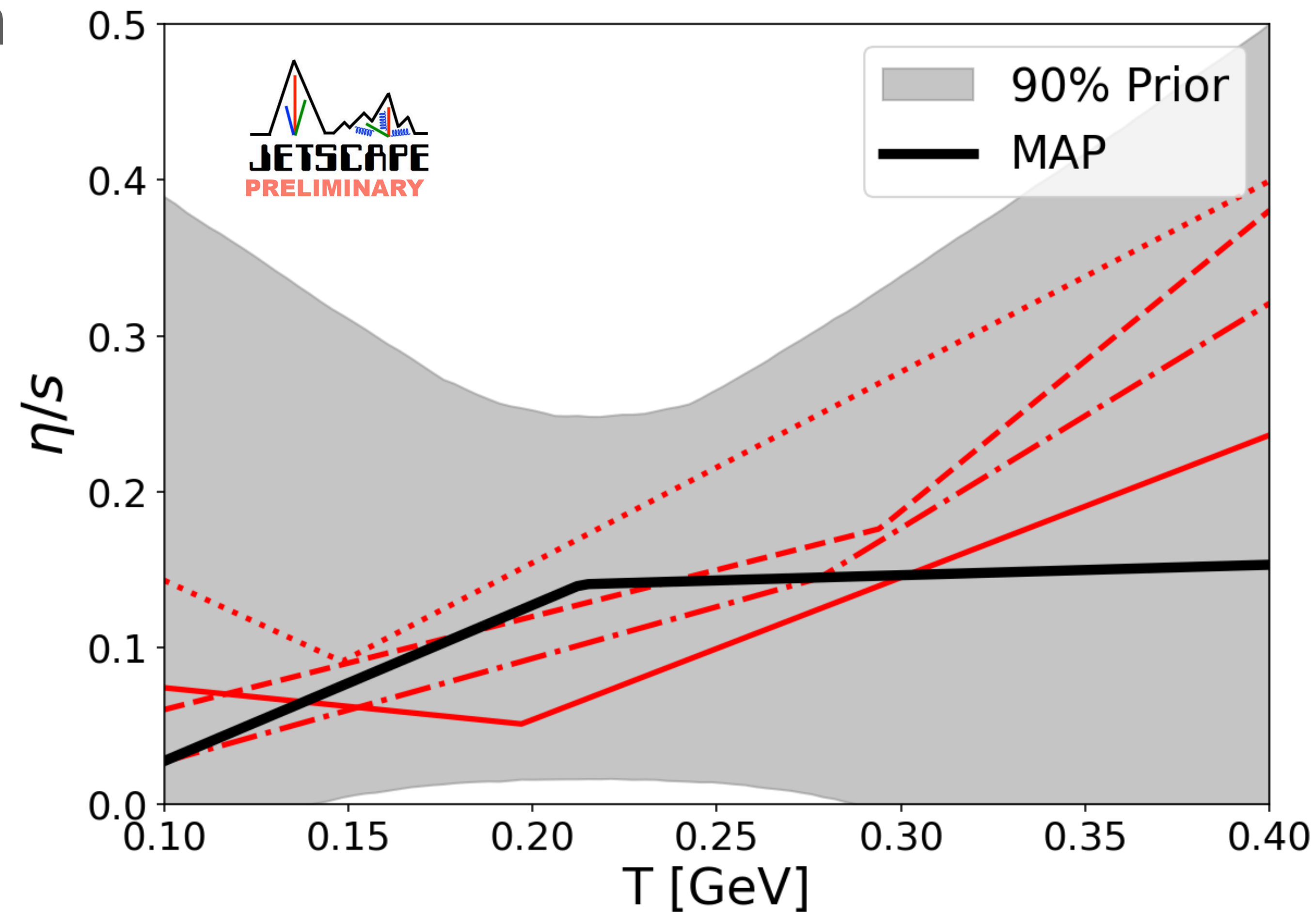
The information content of the MAP vs the full posterior

- The Maximum a Posteriori (MAP) is the highest-likelihood point in parameter space



The information content of the MAP vs the full posterior

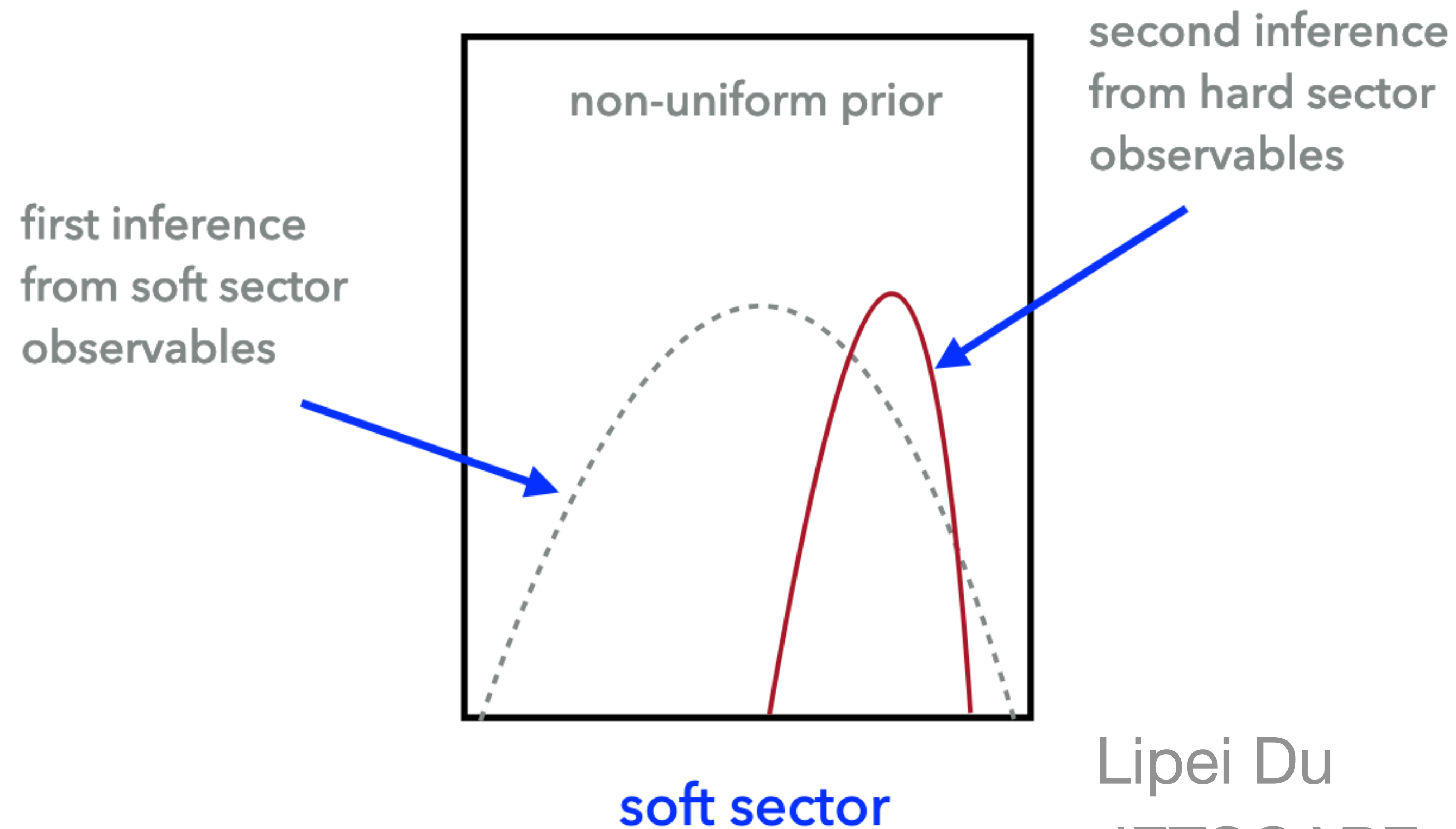
- The Maximum a Posteriori (MAP) is the highest-likelihood point in parameter space
- It does not reflect uncertainties inherent in the full posterior
- A similarly-likely point in parameter space may be physically very different
- To avoid precise but inaccurate predictions, a sample of the posterior should be used whenever feasible



Steps toward including hard observables

- A sequential calibration may be appropriate when:
 - There are no model inconsistencies
 - The datasets do not overlap
- We know the medium properties affect hard observables
- Need to gauge whether sensitivity can overcome experimental and theoretical uncertainties

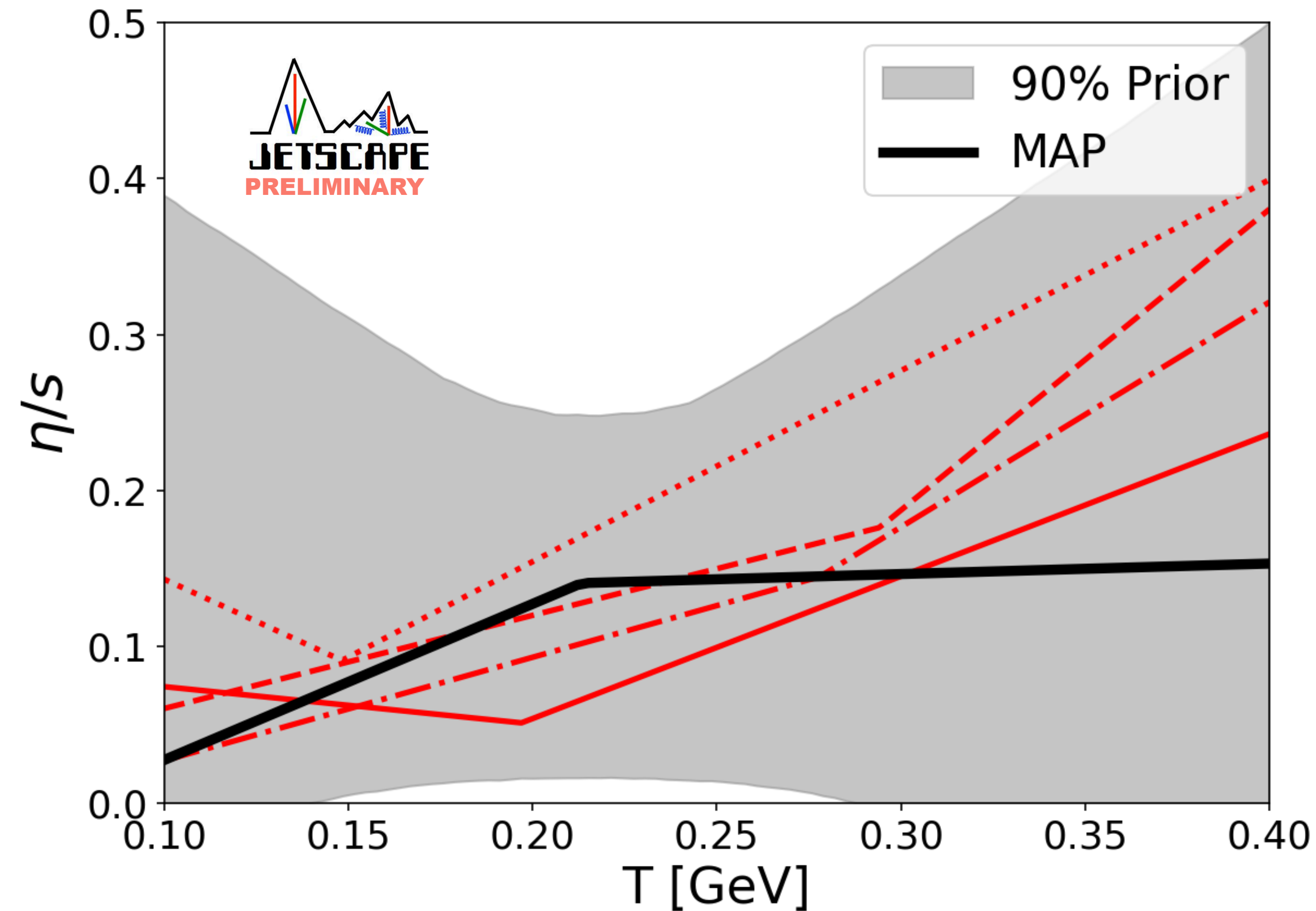
A sequential calibration



Lipei Du
JETSCAPE

Summary

- 3D Bayesian fit to the widest dataset at RHIC top energy may be used as a calibrated medium for future studies, including for jet-medium interaction
- Rapidity-dependent data provides additional constraints for viscosity
- To fully exploit a Bayesian analysis, a sample of the posterior should be used whenever feasible



Acknowledgements



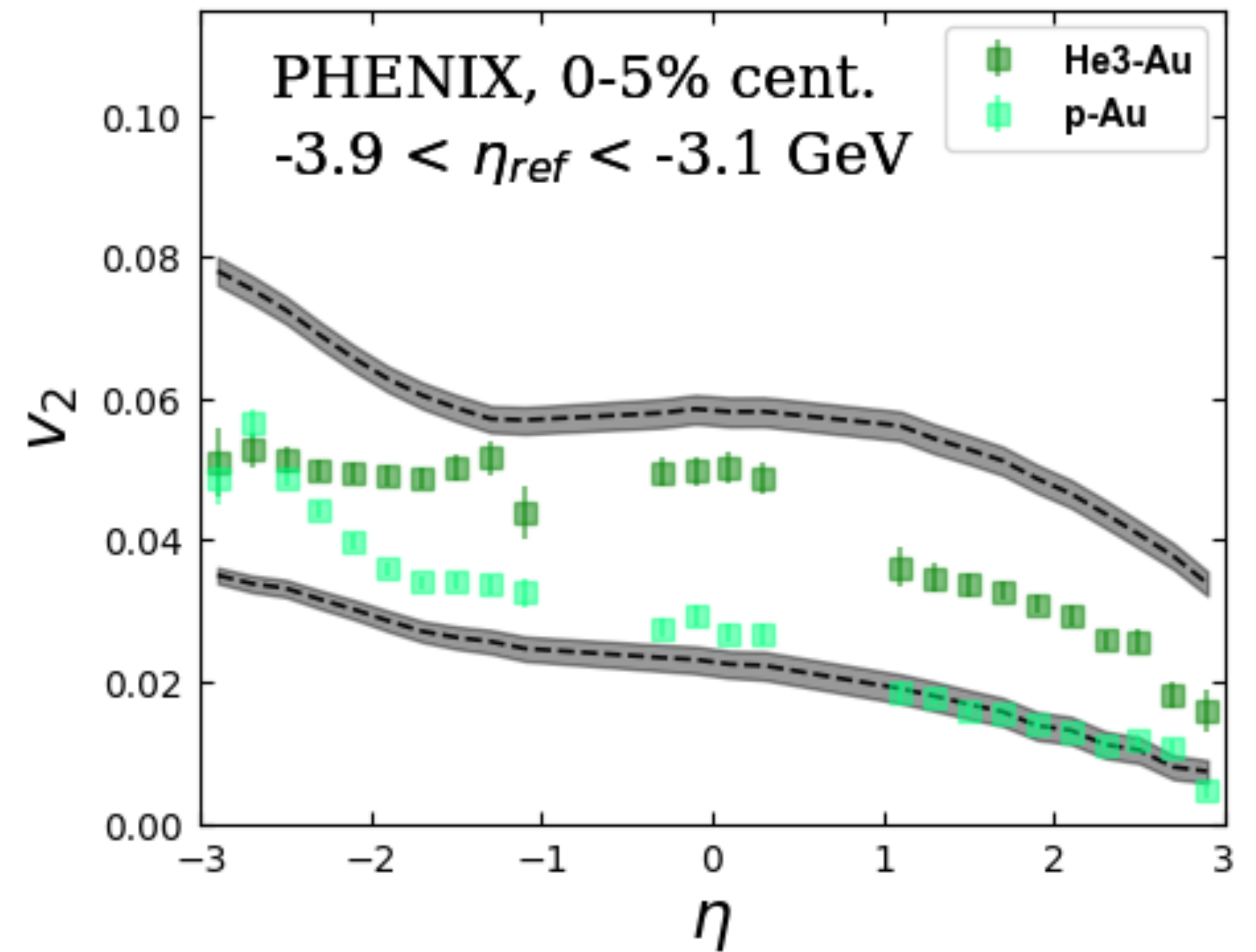
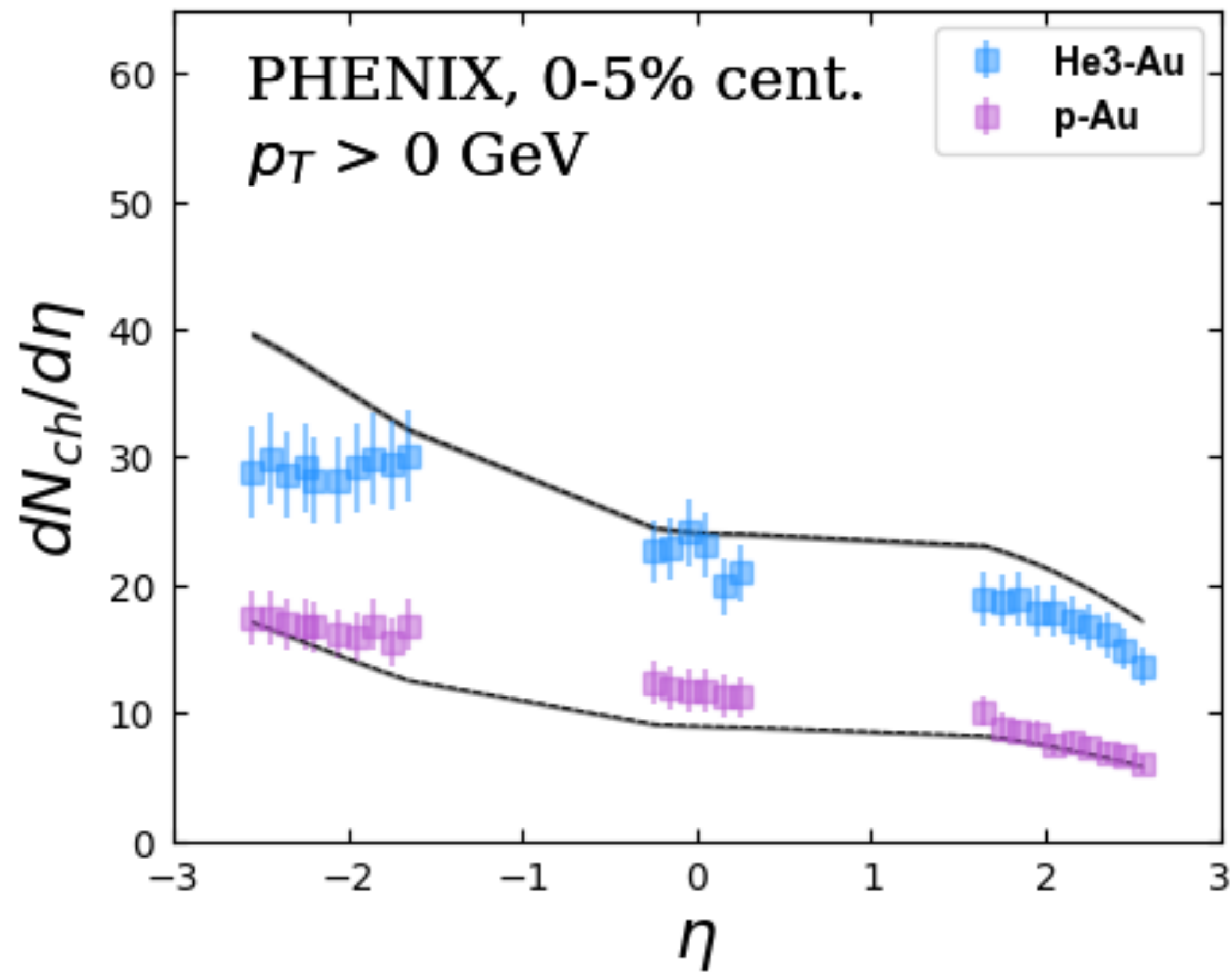
Backup



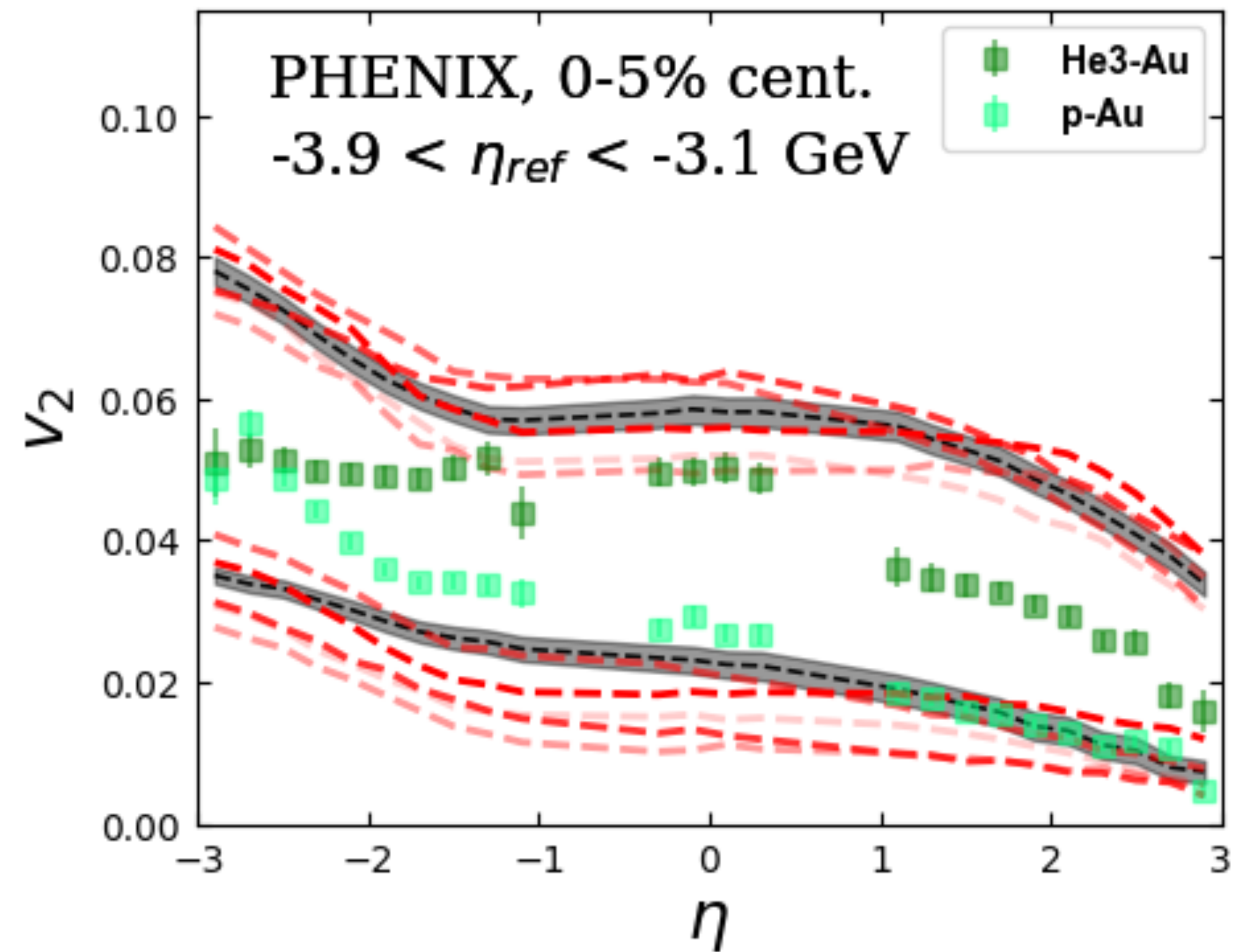
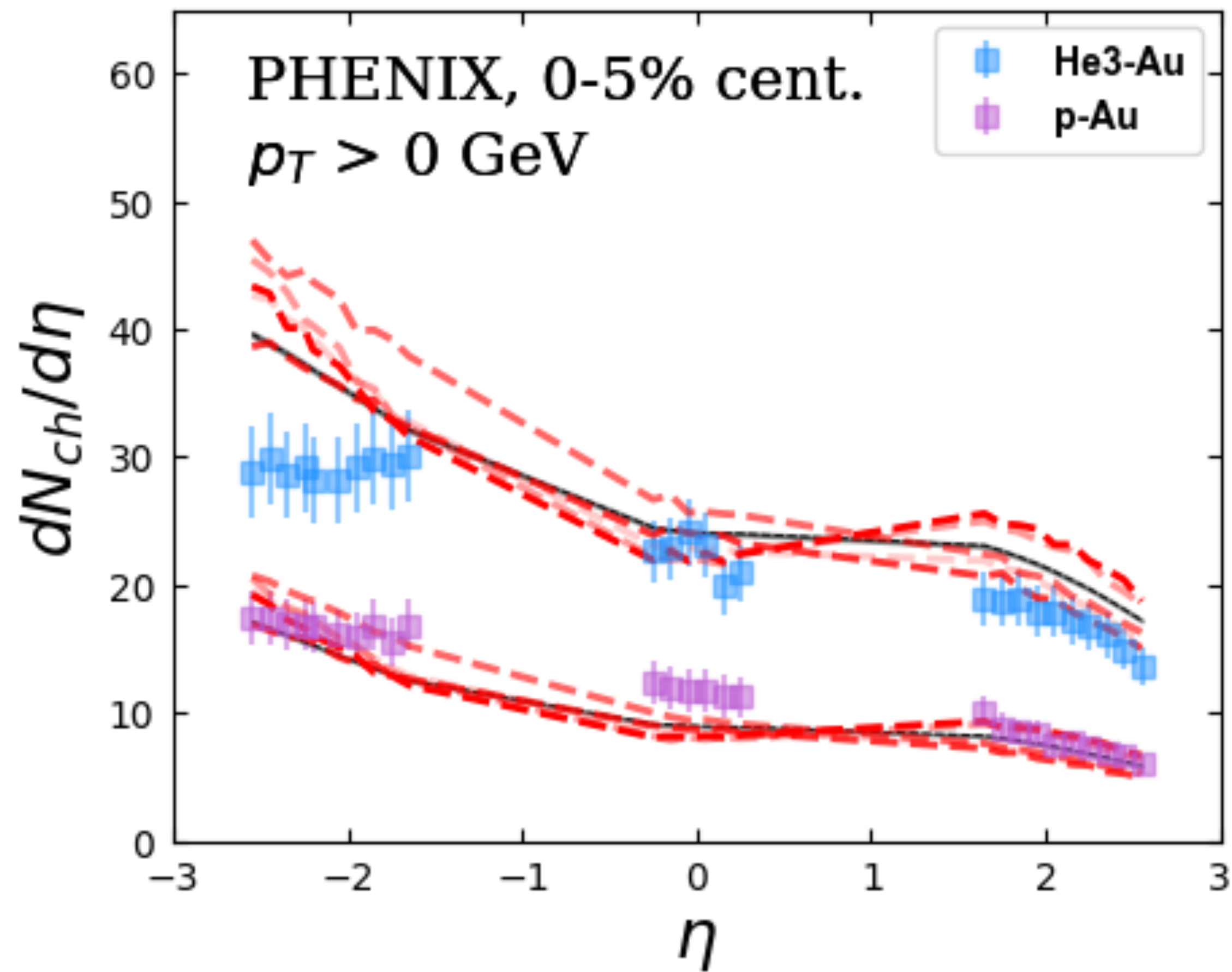
The experimental data

	system	Experiment	cent.bins	data points
$dN_{ch}/d\eta(\eta)$	Au-Au	PHOBOS	11	594
$v_2(\eta)$	Au-Au	PHOBOS	3	48
$v_2(\eta)$	Au-Au	STAR	1	21
$v_2(p_T)$	Au-Au	PHENIX	6	30
$v_2(p_T)$	Au-Au	STAR	7	63
$v_2(\text{cent.})$	Au-Au	STAR	6	6
$v_3(\text{cent.})$	Au-Au	STAR	6	6
$\pi \langle p_T \rangle (\text{cent.})$	Au-Au	STAR	7	7
$k \langle p_T \rangle (\text{cent.})$	Au-Au	STAR	7	7
$k \langle p_T \rangle (\text{cent.})$	Au-Au	PHENIX	8	8
$p \langle p_T \rangle (\text{cent.})$	Au-Au	PHENIX	8	8
$dN_{ch}/d\eta(\eta)$	d-Au	PHOBOS	1	54
$dN_{ch}/d\eta(\eta)$	d-Au	PHENIX	3	102
$v_2(\eta)$	d-Au	PHENIX	1	14
$v_2(p_T)$	d-Au	PHENIX	1	5
$v_2(p_T)$	d-Au	STAR	1	5
Total				978

The information content of the MAP vs the full posterior



The information content of the MAP vs the full posterior



Full likelihood function

$$P(Y_{exp}|\theta) = \frac{1}{\sqrt{2\pi||[\Sigma_{exp} + \Sigma_{sim}(\theta)]||}} \exp \left(-[Y_{exp} - Y_{sim}(\theta)]^T [\Sigma_{exp} + \Sigma_{sim}(\theta)]^{-1} [Y_{exp} - Y_{sim}(\theta)] \right)$$



Model sensitivity to parameters

$$S[x_j] = \Delta_j / \delta$$

x_j - varied parameter

δ - fraction of the latin hypercube training range

$$\Delta_j = \frac{\hat{O}[X'_j] - \hat{O}[X]}{\hat{O}[X]}$$

X - MAP parameters

X'_j - X where x_j is adjusted by an amount proportional to δ

$\hat{O}[\dots]$ - observables evaluated at a given parameter set

